MILITARY SPECIFICATION

SAFETY-ARMING DEVICE

MARK 33 MOD 1

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the MARK 33 MOD 1 safety-arming device, referred to herein as the S-A device.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-H-1168	Helium, Technical.
BB-N-411	Nitrogen, Technical.
TT-I-735	Alcohol, Isopropyl.
UU-T-81	Tags, Shipping and Stock.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1336

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MILITARY

MIL-S-6872	Soldering Process, General Specification for.
MIL-C-45662	Calibration System Requirements.

Naval Air Systems Command
(Code Ident 30003)

WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections.

STANDARDS

FEDERAL

FED-STD-209	Clean Room and Work Station Requirements, Controlled Environment.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-810	Environmental Test Methods.
MS 16997	Screw, Cap, Socket Head and Screw, Cap, Socket Head, Self-locking.
MS 35333	Washer, Lock, Flat-Internal Tooth.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

LD 521540	Booster, Fuze, MARK 38 MOD 1.
LD 521549	Control Console Monitor.

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1778086	Booster Pellet.
2408846	Vibration Fixture, Flight, MARK 33 MOD 0 S-A Device.
SA 2602010	Jolt Fixture.
SA 2602266	Dent Block.
SA 2618609	Warhead Fixture.
DL 917AS3109	Container, Shipping Assembly.
DL 917AS4000	Safety-Arming Device, MARK 33 MOD 1, Data List.

PUBLICATIONS

Code of Federal Regulations

49 CFR 171-179	Transportation.

Naval Air Systems Command (Code Ident 30003)

AD-464	Quality Assurance Provisions on Drawings, Preparation and Use of (Classification of Characteristics).

Naval Weapons Center (Code Ident 12934)

NOLC Report 556	Monitoring the Mechanical Operation of Fuze Safety-Arming Mechanisms.

Naval Ordance Systems Command (Code Ident 10001)

OD-37105	Lot Inspection and Production Summary Reports, Preparation of.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute

ANSI S1.4-1971	Sound Level Meters.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10013.)

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3. REQUIREMENTS.

3.1 Item definition. The MARK 33 MOD 1 S-A device is an electromechanical component of the missile fuzing system and is used to control warhead detonation.

3.1.1 Government-furnished property list. Unless otherwise specified in the contract or purchase order (see 6.2.1), no Government-furnished equipment will be provided.

3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 In-process requirements.

3.2.1.1.1 In-process drawing requirements. Parts and subassemblies of the S-A device shall conform to the applicable drawing requirements of all drawings listed on DL 917AS4000 and all other drawings specified herein.

3.2.1.1.2 Electrical requirements. The escapement-mechanism assembly (Drawing 917AS4001), prior to installing any detonators, shall meet the resistance or capacitance limits specified for the indicated test points (see Table I). Tests No. 1 through 12 pertain to the mechanism in the safe condition and 13 through 17 in the armed condition. Test point letters correspond to pin numbers on S-A connector J1 (see Drawing 917AS4027).

3.2.1.1.3 Launch latch functioning. The launch latch shall function as follows:

(a) The launch latch shall withdraw completely from the setback weight.

(b) With the mechanism at rest, the launch latch shall lock the setback weight in the forward position when voltage is removed from the solenoid.

3.2.1.1.4 Seal. There shall be no visible evidence of seal leakage (see 4.2.1.5(a)). The seal shall not leak at a rate greater than 5×10^{-5} standard cubic centimeters per second (cm^3/s) (see 4.2.1.5(b)).

3.2.1.2 End item requirements. The following requirements shall be met by the sealed, pressurized, and loaded S-A device during the end item quality conformance inspections.

3.2.1.2.1 End item drawing requirements. The S-A device shall conform to the applicable drawing requirements of all drawings listed on DL 917AS4000 and all other drawings specified herein.

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TABLE I. Escapement mechanism assembly electrical requirements.

Test No.	Test points		Limits		Unit
	From (+)	To (-)	Minimum	Maximum	
1	C	D, E11, E12 frame[a]	100.00	- - -	Mohm
2	D	E11, E12 frame[a]	100.00	- - -	Mohm
3	E11	E12, frame[a]	100.00	- - -	Mohm
4	E12	Frame	100.00	- - -	Mohm
5	A	C	100.00	- - -	Mohm
6	A	Frame	100.00	- - -	Mohm
7	D	F	10.00	12.50	kohm
8	D	H	27.00	32.00	ohm
9	D	E8	1.80	2.20	kohm
10	A	B	13.50	16.50	kohm
11	A	F	4.00	11.00	Mohm
12	A	E8	16.20	- - -	µF
13	E12	C	- - -	0.15	ohm
14	D	E11	- - -	0.15	ohm
15	C	E8	- - -	0.30	ohm
16	E5	F	19.80	24.40	kohm
17	D	F	3.75	4.59	kohm

[a]If test results are below minimum specified with all pins connected on negative side, then check individual pin to pin designated + to verify if resistance is within the limits specified.

3.2.1.2.2 Laboratory arming. The S-A device shall meet the following requirements when subjected to the conditions specified in Section 4. One set of time limits specified below applies to the acoustic method of measuring arming time (see 4.2.2.2.1); the other applies to the telemeter (TLM) method of measurement (see 4.2.2.2.2).

(a) The S-A device shall not arm (see 4.2.2.2.1(a) or 4.2.2.2.2(a)).

(b) The applicable time limits indicated in Table II shall be met (see 4.2.2.2.1(b) or 4.2.2.2.2(b)).

(c) The applicable time limits indicated in Table II shall be met (see 4.2.2.2.1(c) or 4.2.2.2.2(c)).

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(d) The applicable time limits indicated in Table II shall be met (see 4.2.2.2.1(d) or 4.2.2.2.2(d)).

(e) The S-A device shall arm. (Time limits not applicable.) (See 4.2.2.2.1(e) or 4.2.2.2.2(e).)

TABLE II. Time limits, seconds.

Acoustic method	Maximum	Minimum	TLM method	Maximum	Minimum
4.2.2.2.1(b)	2.188[a]	1.495[a]	4.2.2.2.2(b)	1.727[a]	1.495[a]
	2.277[b]	1.410[b]		1.797[b]	1.410[b]
4.2.2.2.1(c)	3.461[a]	2.365[a]	4.2.2.2.2(c)	2.731[a]	2.365[a]
	3.600[b]	2.229[b]		2.841[b]	2.229[b]
4.2.2.2.1(d)	4.468[a]	3.053[a]	4.2.2.2.2(d)	3.526[a]	3.053[a]
	4.648[b]	2.878[b]		3.668[b]	2.878[b]

[a]Required time limits at the contractor's facility.

[b]Required time limits after leaving contractor's facility.

3.2.1.2.2.1 Switch functioning. Switch S2d (which is normally closed when the S-A device is in the safe condition) shall open three times during a test where *arming takes place and shall remain open after the third opening.* Switch S2e (which is normally open when the S-A device is in the safe condition) shall remain open after acceleration begins and shall remain open until the third opening of switch S2d. Switch S2e shall then close as arming takes place. When the S-A device is required to be in the safe condition (at rest with the solenoid voltage removed), the setback weight shall be locked in the forward position by the launch latch. (No external force, e.g., tapping or shaking, shall be employed to ensure conformance with this requirement.)

3.2.1.2.3 Electrical resistance. The S-A device shall meet the following resistance limits. Test point letters correspond to pin numbers on S-A connector J1 (see Drawing 917AS4027).

3.2.1.2.3.1 Resistance (safe condition). With the S-A device in the safe condition, the resistance limits of Table III shall be met.

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TABLE III. Resistance limits (safe condition).

Test No.	Test points		Resistance limits		Unit
	From (+)	To (-)	Minimum	Maximum	
*1	C	Frame, D[a]	100	. . .	Mohm
*2	D	Frame	100	. . .	Mohm
*3	A	C	100	. . .	Mohm
4	D	F	10.0	12.5	kohm
5	D	H	27.0	32.0	ohm
6	A	B	13.5	16.5	kohm
7	A	F	4.0	11.0	Mohm

[a]If test results are below minimum specified with D connected to frame on negative side, then check frame to C and D to C individually to verify if resistance is within limits specified.

*Resistance shall be met with +500 ±50 volts direct current (Vdc) applied.

3.2.1.2.3.2 Resistance (armed condition). With the S-A device in the armed condition, the resistance limits of Table IV shall be met:

TABLE IV. Resistance limits (armed condition).

Test No.	Test points		Resistance limits		Unit
	From (+)	To (-)	Minimum	Maximum	
1	D	C	1.0	4.0	ohm
2	D	H	27.0	32.0	ohm
3	D	F	3.8	4.6	kohm

3.2.1.2.3.3 Detonator circuit continuity determination. With the S-A device in the armed condition, the detonator circuit, test points C(-) to D(+), shall indicate continuous continuity.

3.2.1.2.4 Seal.

3.2.1.2.4.1 At place of manufacture. When tested in accordance with 4.2.2.4.1 at the place of manufacture, the S-A device shall not leak at a rate greater than 5×10^{-5} standard cm^3/s.

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3.2.1.2.4.2 After shipment. After shipment from the contractor's facility, the S-A device shall meet the following requirements:

(a) When immersion-tested, there shall be no visible evidence of leakage (see 4.2.2.4.2(a)).

(b) The S-A device shall not leak at a rate greater than 5×10^{-4} standard cm^3/s (see 4.2.2.4.2(b)).

3.2.1.2.5 Safety determination. Upon satisfactory compliance with all of the requirements of 3.2.1.2.1 through 3.2.1.2.4.1, and just prior to packaging for shipment, each S-A device shall meet the following requirements.

(a) A silver (bare metal) letter "S" on a green background, visible through inspection window.

(b) No evidence of red indicator as viewed through launch latch viewing window.

(c) No explosive component of the S-A device shall initiate.

Upon successful completion of the safety test, the contractor shall fill in the safety inspection tag (Figure 1).

3.2.1.2.6 Visual inspection. The S-A device shall meet the following requirements:

(a) No external damage shall be evident; i.e., cracked or broken windows, dents, or other marks which may indicate damage which could interfere with functioning or fit at the next assembly.

(b) Letter "S" in the aft window and no red visible in the forward window.

(c) A band of 3/4-inch-wide yellow paint around the exterior of the housing (to indicate a loaded unit).

(d) Connector dust cap installed.

(e) Safety-inspection tag (Figure 1) attached through mounting flange hole as shown on assembly drawing and properly filled in.

(f) Lead seal crimped onto safety-inspection tag wires (indicates safety determination of 4.2.2.5 has been performed successfully).

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(g) All signatures are on safety-inspection tag as required.

(h) All printed wiring boards and solder terminations shall meet the requirements of WS-6536.

3.2.1.2.7 Environmental requirements. The S-A device shall meet the environmental requirements specified below. The term "operable," as used in this specification or in referenced documents, is defined as the ability of the S-A device to meet the requirements of any tests to be performed following an environmental test in Figures 2 and 3.

3.2.1.2.7.1 Jolt. The S-A device shall meet the acceptance criteria of MIL-STD-331, Test 101, Procedure II, and after completing the test of 4.2.2.2.1(b) and 4.2.2.2.2(b) shall either fail to arm or shall arm in not less than the minimum time shown in 3.2.1.2.2(b).

3.2.1.2.7.2 Forty-foot drop. The S-A device shall meet the acceptance criteria of MIL-STD-331, Test 103, Procedure 1. To satisfy the requirements of MIL-STD-331, the disassembled S-A device shall display the following conditions:

(a) No explosive component beyond the detonators shall have initiated.

(b) No loose explosive shall be present in the housing.

(c) The control rotor pin on the control rotor shall not be broken.

(d) The lead rotor shall be in the safe position, secured so that it cannot be turned to the armed position with finger pressure.

(e) The setback weight shall be in the full forward position and locked by the solenoid latch, and the bias springs shall be in place.

3.2.1.2.7.3 Transportation vibration. The S-A device shall meet the acceptance criteria of MIL-STD-331, Test 401, Procedure II.

3.2.1.2.7.4 Humidity. The S-A device shall remain safe and operable after testing.

3.2.1.2.7.5 Flight vibration before arming. The S-A device shall remain safe during, and be operable after, testing.

3.2.1.2.7.6 Flight vibration after arming. The S-A device shall remain armed, explosive components shall not initiate, and switches or associated S-A device components shall not fail in any way which may interfere with initiation of the firing train.

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3.2.1.2.8 Explosive train requirements.

3.2.1.2.8.1 Static detonator safety. There shall be no detonation, burning, charring, scorching, melting, or rupturing of the MARK 8 MOD 0 (LD 480140) or MARK 16 MOD 0 (LD 521882) explosive leads. Smudging of the surfaces of the explosive components mentioned, due to deposits resulting from firing of the MARK 71 MOD 0 electric detonators (LD 486252), is permitted. Rupturing of the S-A device housing or tape is not permitted.

3.2.1.2.8.2 Explosive train output. The explosive train of the S-A device shall produce a dent in the dent block (Drawing SA 2602266) having a depth of at least 0.050 inch.

3.3 Design and construction.

3.3.1 Production drawings. The S-A device shall be fabricated and assembled in accordance with the drawings, parts list, and other documents listed on DL 917AS4000.

3.3.2 Standards of manufacture.

3.3.2.1 Assembly conditions. The S-A device shall be assembled in a careful, neat, and orderly manner to avoid inclusion of dust or other foreign matter and to prevent damage to parts. The working environment shall meet the levels and requirements specified in Table V. In addition to the requirements of Table V, devices shall be kept in clean, covered plastic containers when not in use. No lubricants of any kind shall be used on any part of the device after final cleaning of components except as specifically provided for on the drawings. The soldering techniques used in the manufacture of the S-A device shall meet the requirements of WS-6536 for electrical connections within. For mechanical soldering, MIL-S-6872 shall apply.

3.3.3 Workmanship. Workmanship during the manufacture of the S-A device shall meet the requirements of WS-6536 and MIL-STD-454, Requirement 9. The requirement for first-class workmanship shall not be limited solely to the above standards, and shall not preclude the use of time- and labor-saving techniques commensurate with the expendable nature of this S-A device, provided reliability and safety will in no way be jeopardized.

3.4 First article.

3.4.1 Preproduction sample. Unless otherwise specified in the contract or purchase order (see 6.2.1), a preproduction sample of 45 S-A devices shall be delivered to the testing activity designated in the contract or purchase order

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TABLE V. Minimum standards for working environments.

Type of Work	Cleanliness Note 1	Lighting Foot Candles Note 2	Air Temperature Note 3	Relative Humidity Note 4	Dust Control Note 5	Ventilation or Exhaust Note 6	Noise Note 7 dbA	Habitat
Machine Shop								
Tolerance to 0.01	Class D	70	60–100°F	U	Class D	Note 6	90 dbA	NOTE 8
Tolerance to 0.001	Class C	100	65–95°F	U	Class C	Note 6	80 dbA	NOTE 8
Tolerance to 0.0001	Class B	200	65–85°F	70% Max.	Class C	Note 6	80 dbA	NOTE 8
Tolerance finer than 0.0001	Class BA	500	65–75°F	60% Max.	Class B	Note 6	75 dbA	Note 9
Propellant Chemical and Potting Areas	Class B	100	65–95°F	60% Max.	Class B	Note 6	80 dbA	Note 9
Plating and Heat Treating	Class C	70	55–105°F	U	Class C	Note 6	90 dbA	Note 8
Electronic Assembly	Class B	100	65–95°F	30–70%	Class B	Note 6	75 dbA	Note 9
Mechanical Assembly								
Tolerance to 0.05	Class C	100	65–95°F	U	Class C	Note 6	75 dbA	Note 8
Tolerance to 0.001	Class B	100	65–95°F	U	Class C	Note 6	75 dbA	Note 8
Tolerance to 0.0001	Class BA	200	65–85°F	70% Max.	Class B	Not Req	70 dbA	Note 9
Tolerance finer than 0.0001	Class A	500	65–75°F	30–50%	Class A	Not Req	70 dbA	Note 9
Inspection Stations	Equal to product inspected							

1. Cleanliness Definitions

   a. Class D – – Daily Cleanup: Removal of scrap, clean up all spilled oil, etc.

   b. Class C – – Prompt Cleanup: Scrap, oil and residue shall not be allowed to accumulate.

   c. Class B – – Prompt Cleanup: Oil, residue, spilled chemicals removed immediately. Floors, walls and work area shall have hard, grease resistant, easily cleaned surfaces. Cleanup of equipment and area shall be accomplished after each 24 hours of operation.

   d. Class BA – – Oil, residue, spilled chemicals, and any foreign material which might develop in the area shall be cleaned immediately. Floors, walls, furniture and work areas shall have hard, grease resistant, easily cleaned surfaces. Materials to be worked on shall be cleaned of all foreign material prior to acceptance into this area. Flushing or cutting fluids shall be filtered to remove contamination above 30.0 microns. Chips, particles and dust generated during any and all mechanical operation shall be removed during the operation by vacuum. Clean lint-free smocks are required for operators and inspectors. Complete cleaning and wipe down of equipment, tools, fixtures and the area shall be accomplished at the end of each 8 hours of operation or each 24 hour period.

   e. Class A – – Cleanliness controlled in accordance with FED–STD–209, Class 100,000.

2. Lighting

   Indicated values are minimum light intensity values at the work station. Supplemental lighting shall be used when necessary to improve precision and minimize operator fatigue, but brightness ratios within the operators' field-of-view shall not exceed 10 to 1.

3. Air Temperature

   Designated temperature limits are the daily average temperature measurements taken in proximity of the work stations. Ambient temperature shall be between 65°F and 85°F.

4. Relative Humidity

   Designated relative humidity shall be as measured at room ambient temperature. "U" indicates uncontrolled relative humidity.

5. Dust Control Definitions

   a. Class D – – No dust control required.

   b. Class C – – Outside air shall be filtered to remove dust particles. Type of filter is unspecified.

   c. Class B – – Outside and recirculated air shall be filtered to remove dust particles. Filter rating shall be 10 micron maximum.

   d. Class A – – Dust control shall be in accordance with FED–STD–209, Class 100,000.

6. Ventilation or Exhaust

   Forced ventilation or exhaust shall be provided whenever required to minimize operator fatigue.

7. Noise

   Noise is defined as the average sound level existing at the work station during normal operation when measured with a standard sound-level meter as specified in ANSI S1.4–1971 "A" weighted. Work stations for control of automatic equipment-operations, which requires operator loading and unloading only, may exceed the noise limits specified by 10 dbA but not to exceed 90 dbA. Inspection and test work stations shall not exceed the values specified. Work areas for mechanical assembly of tolerances finer than 0.001 which utilize laminar flow benches may exceed by 5 dbA the value specified.

8. Habitat

   No eating, drinking or personal grooming is allowed in these work areas. Drinking fountains are permissible.

9. Habitat

   No eating, drinking, smoking or personal grooming is allowed in these work areas. Drinking fountains are permissible.

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for preproduction testing (see 6.2.1). The preproduction sample shall be manufactured using the same methods, materials, and processes proposed for production. Further manufacture of the S-A device prior to approval of the preproduction sample shall be at the risk of the contractor. Any change in manufacturing conditions, methods, or procedures after approval of the contractor's preproduction sample may, at the discretion of the procuring activity, be cause for the contractor to submit a new preproduction sample.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. Inspections and tests to be performed on the S-A device shall be classified as follows:

(a) Preproduction inspections. Preproduction inspections and tests shall be as specified in 4.1.2.1 and Figure 2.

(b) Quality conformance inspections. Quality conformance inspections and tests shall be as specified in 4.2 and Figures 3, 4 and 5.

4.1.1.1 Test equipment.

4.1.1.1.1 Standard test equipment. The contractor shall provide and maintain the test equipment necessary to ensure that the parts, subassemblies, and final product will meet the requirements of this specification. Test equipment, calibration programs, and operation and maintenance procedures shall be subject to disapproval by the procuring activity. Unless otherwise specified, any error introduced by the test equipment shall not exceed 10 percent of the tolerance of the requirement being measured. The test equipment shall be controlled and calibrated in accordance with MIL-C-45662.

4.1.1.1.2 Special test equipment. Unless otherwise specified by the procuring activity, the following special test equipment is required to test the S-A device:

(a) Vibration fixture (Drawing 2408846).

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(b) Jolt fixture (Drawing SA 2602010).

(c) Warhead fixture (Drawing SA 2618609).

(d) Dent block (Drawing SA 2602266).

4.1.1.1.3 Gages. The contractor shall provide all necessary gages to ensure that the S-A device will meet the dimensional requirements shown on the applicable drawings.

4.1.2 Special test and examinations.

4.1.2.1 Preproduction inspections. The preproduction sample shall be subjected to the tests and inspections specified in Figure 2 in the sequence shown therein. The inspections below the dashed line of Figure 2 shall be performed at a facility designated by the procuring activity (see 6.2.1).

4.1.2.1.1 Preproduction inspection acceptance criteria. Failure of any S-A device to pass any of the tests or examinations of Figure 2 shall result in rejection of the preproduction sample.

4.1.2.2 Quality conformance inspections. Quality conformance inspections shall consist of the following tests:

(a) In-process inspections. In-process examinations and tests shall be performed during manufacture, prior to formation of the inspection lot, in accordance with 4.2.1. The inspection sequence shall be in accordance with Figure 3.

(b) End-item inspections. End-item (lot acceptance) tests and examinations shall be performed on all inspection lots in accordance with 4.2.2. The inspection sequence shall be in accordance with Figure 4 (nondestructive tests) and Figure 5 (destructive tests). The tests of Figure 5 shall be performed at a facility designated by the procuring activity (see 6.2.1).

4.1.2.2.1 Quality conformance inspection acceptance criteria. Failure of any S-A device to conform with the requirements of the contract or purchase order, applicable drawings, or this specification shall result in rejection of the S-A device. If the number of defectives recorded during the tests of 4.2.2.1 reaches the rejection criteria of the specified sampling plan, the lot sample shall be inspected 100 percent for such defects and the defective units shall be rejected on an individual basis. The lot shall be rejected (see 6.2.1) when one of the following occurs:

(a) When the number of defectives in the sample units specified in Figure 4 reach the rejection level specified in MIL-STD-105, Double Sampling Plan, sample size code letter k, Acceptable Quality Level (AQL) 0.40.

(b) When one or more defectives are found in the 15 sample units during the tests of Figure 5.

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4.1.2.2.2 Sampling. Unless otherwise specified herein, sampling shall be in accordance with MIL-STD-105, Inspection Level II, Single Sampling Plan, AQL 1.0, Normal Inspection.

4.1.2.2.3 Inspection lot formation. Unless otherwise specified in the contract or purchase order (see 6.2.1), an inspection lot is defined as not less than 160 nor more than 500 S-A devices, including those devices required for destructive testing (see Figure 5). All units shall be alike in construction and place of manufacture. The term "inspection lot" as used in this specification is defined as a number of units grouped together for inspection purposes only, and this term has no relation to, nor is it intended to conflict with, the term "lot" as may be used in the contract or purchase order with regard to the rate of production or delivery. All S-A devices shall be assigned a Government serial number (see 6.2.1). After assignment of a Government serial number to each S-A device, the inspection lot shall be formed and identified with a lot number. Prior to forming the inspection lot, the contractor may subject S-A devices to preliminary testing in accordance with 4.2.2. However, the environmental tests of 4.2.2.7 through 4.2.2.8.2 are considered destructive and S-A devices subjected to any of the tests of 4.2.2.7 through 4.2.2.8.2 shall not be included in the inspection lot. After the inspection lot has been identified, the entire lot shall be tested in accordance with 4.2.2.

4.1.2.3 Inspection records. The results of all the test and examinations given in Figures 2 through 5 shall be maintained by the contractor (see 6.2.1) and recorded on the inspection record shown in Figure 6 where specified herein. The inspection record for each S-A device shall be submitted to the procuring activity (see 6.2.2) in accordance with Section 5. Lot inspection reports in accordance with OD-37105 shall be prepared by the contractor for delivery to the procuring activity (see 6.2.2).

4.1.2.4 Production summary report. Production summary reports in accordance with OD-37105 shall be prepared by the contractor for submission to the procuring activity (see 6.2.2).

4.2 Test procedures, provisions, and equipment.

4.2.1 In-process inspections. The following tests and inspections shall be performed in the sequence shown in Figure 3 during manufacture of the S-A device. Parts and subassemblies found to be defective during the tests of 4.2.1.1 through 4.2.1.5 shall not be used in further assembly of the S-A device.

4.2.1.1 Drawing inspections (nondestructive). Each production lot of parts or subassemblies shall be sampled and inspected for the classification of characteristics on the applicable drawing in accordance with AD-464. Where acceptance criteria are not indicated herein, they will be found on the referenced drawing. If a defect is observed in a sample from a particular lot, that lot shall be inspected 100 percent for the defect observed to determine whether more of these defects exist. After completion of the inspection, acceptable units may be returned to the lot for further assembly.

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	REQUIREMENT	DRAWING
CRITICAL - Sample as specified on referenced drawing:		
1.	Verify that escapement gear and pinion assembly meet the push-off test requirements.	917AS4015
2.	Verify that No. 1 gear and pinion assembly meets the push-off test requirements.	917AS4029
3.	Verify that pallet-lever assembly pins meet the push-off test requirements.	917AS4018
4.	Verify that pallet-lever assembly shaft meets the push-off test requirements.	917AS4018
5.	Verify that No. 2 gear and pinion assembly meets the push-off test requirements.	917AS4032
6.	Verify that lead rotor assembly shaft meets the push-off test requirements.	917AS4017
7.	Verify that control rotor assembly shaft meets the push-off test requirements.	917AS4016
8.	Verify that escapement sideplate and pillar assemblies meet the push-off test requirements.	917AS4003
9.	Verify that setback weight meets the tensile force requirements.	917AS4006
10.	Verify proper color and location of red-orange paint (Drawing 917AS4076-1).	917AS4006
11.	Verify proper color and location of green paint (Drawing 917AS4076-2).	917AS4017
12.	Verify proper color and location of red-orange paint (Drawing 917AS4076-1).	917AS4017

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	REQUIREMENT	DRAWING
MAJOR - Use MIL-STD-105, AQL 0.40, Table IIA, for inspections 101 through 114.		
101.	Verify 0.0889 inch BASIC between two hole centers.	917AS4044
102.	Verify 0.200 inch BASIC between two hole centers.	917AS4044
103.	Verify 0.0889 inch BASIC between two hole centers.	917AS4045
104.	Verify 0.200 inch BASIC between two hole centers.	917AS4045
105.	Verify 0.0264 inch BASIC between two hole centers.	917AS4044
106.	Verify 0.375 inch BASIC between two hole centers.	917AS4044
107.	Verify 0.0264 inch BASIC between two hole centers.	917AS4045
108.	Verify 0.375 inch BASIC between two hole centers.	917AS4045
109.	Verify gear tooth configuration of escapement gear. (Comparator chart may be used.)	917AS4046
110.	Verify 0.010 inch minimum tip thickness of escapement gear after deburring.	917AS4046
111.	Verify secure attachment (closed ends) of solenoid spring (Drawing 917AS4074) to rotor assembly and solenoid housing.	917AS4009
112.	Verify that spring pin (MS171492) is located within the 0.037 inch maximum tolerance specified.	917AS4016
113.	Verify 36 ±5 inch-ounce torque of four screws (Drawing 917AS4041).	917AS4001
114.	Verify that eccentric weight of lead rotor meets the push-off test requirements.	917AS4017
CRITICAL - Certification (see 6.6 for definition) is required for inspections 13 through 29:		
13.	Verify material composition of No. 1 gear.	917AS4031
14.	Verify heat treatment of No. 1 gear.	917AS4031
15.	Verify material composition of No. 2 gear.	917AS4034

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	REQUIREMENT	DRAWING
16.	Verify heat treatment of No. 2 gear.	917AS4034
17.	Verify material composition of No. 1 pinion.	917AS4030
18.	Verify heat treatment of No. 1 pinion.	917AS4030
19.	Verify material composition of No. 2 pinion.	917AS4033
20.	Verify heat treatment of No. 2 pinion.	917AS4033
21.	Verify material composition of housing.	917AS4023
22.	Verify heat treatment after forming.	917AS4023
23.	Verify material composition of escapement pinion.	917AS4025
24.	Verify heat treatment of escapement pinion.	917AS4025
25.	Verify material composition of escapement gear.	917AS4046
26.	Verify heat treatment of escapement gear material.	917AS4046
27.	Verify material composition of setback weight activating pin.	917AS4062
28.	Verify heat treatment of setback weight activating pin.	917AS4062
29.	Verify material composition of lever pin.	917AS4103

MAJOR - Use MIL-STD-105, AQL 1.0, Table IIA for inspections 115 through 118 (comparator chart may be used):

115.	Verify gear tooth configuration of No. 1 gear.	917AS4031
116.	Verify gear tooth configuration of No. 2 gear.	917AS4034
117.	Verify gear tooth configuration of No. 1 pinion.	917AS4030
118.	Verify gear tooth configuration of No. 2 pinion.	917AS4033

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	REQUIREMENT	DRAWING
For inspections 119 through 133, 100 percent inspection is required:		
119.	Verify that solenoid pole and coil assembly have no perceptible end play.	917AS4009
120.	Verify that solenoid rotor assembly is crimped 360 degrees into solenoid housing assembly.	917AS4009
121.	Verify that solenoid tubes are crimped to bearing retainer 360 degrees and have no sharp edges.	917AS4009
122.	Verify that MARK 8 MOD 0 lead is correctly installed and staked and that the stake does not perforate or deform the lead in any way.	917AS4017
123.	Verify that setback weight runs free and latches in both forward and aft position.	917AS4002
124.	Verify presence of two MARK 71 MOD 0 detonators.	917AS4001
125.	Verify that detonator leads are soldered in place.	917AS4001
126.	Verify that detonator leads are insulated with tubing.	917AS4001
127.	Verify that ball bearing is crimped 360 degrees or four-point-minimum-staked into solenoid housing assembly.	917AS4013
128.	Verify that MARK 16 MOD 0 lead is correctly installed and staked and that the stake does not perforate or deform the lead in any way.	917AS4066
129.	Verify that latch lock is pinned by spring pin (MS171431).	917AS4012
130.	Verify presence of spring pin (MS171433) in solenoid rotor assembly.	917AS4012
131.	Verify that retainer nut (Drawing 917AS4059) has been tightened as specified.	917AS4000
132.	Verify that evacuation tube is crimped, soldered, has no cracks, and is of proper length.	917AS4000
133.	Verify that the seal washer assembly, Drawing 917AS4057, has been properly soldered in place.	917AS4000

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4.2.1.2 Drawing inspections (destructive). Either sampling plan specified on the referenced drawing may be used. After completion of the inspection, tested units shall be considered destroyed and shall not be used for further assembly. Accept-reject criteria are specified on the referenced drawings.

	REQUIREMENT	DRAWING
CRITICAL - Sample as specified on referenced drawing:		
30.	Verify that escapement gear and pinion assembly meets the push-off test requirements.	917AS4015
31.	Verify that No. 1 gear and pinion assembly meets the the push-off test requirements.	917AS4029
32.	Verify that pallet-lever assembly pins meet the push-off test requirements.	917AS4018
33.	Verify that pallet-lever assembly shaft meets the push-off test requirements.	917AS4018
34.	Verify that No. 2 gear and pinion assembly meets the push-off test requirements.	917AS4032
35.	Verify that lead rotor assembly shaft meets the push-off test requirements.	917AS4017
36.	Verify that control rotor assembly shaft meets the push-off test requirements.	917AS4016
37.	Verify that escapement sideplate and pillar assemblies meet the push-off test requirements.	917AS4003

4.2.1.3 Electrical tests. Electrical measurements shall be made on the escapement-mechanism assembly (Drawing 917AS4001) using test equipment which meets the requirements of 4.1.1.1. The resistance between the test points indicated in 3.2.1.1.2 shall be measured to determine conformance with the requirements listed in the same order in 3.2.1.1.2. Tests No. 1 through 6 of Table I shall be conducted using +750 ±75 Vdc.

WARNING

Before starting test, verify that no detonators have been installed.

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4.2.1.4 Launch latch functioning. With the S-A device in the safe condition, determine conformance with 3.2.1.1.3 by applying +18 +0, -1 Vdc to pins H and D of the connector and then removing voltage while observing action of launch latch.

4.2.1.5 Seal. To determine conformance with 3.2.1.1.4, the S-A device shall be subjected to both of the following tests:

(a) After solder-sealing the case of the S-A device, and immediately before sealing the evacuation tube, pressurize the assembly internally to 35 pounds per square inch absolute (psia) using a mixture of 90 percent nitrogen in accordance with BB-N-411, Type I, Class 1, Grade B and 10 percent helium in accordance with BB-H-1168, Type I, Grade B. Maintain this pressure for 5 minutes minimum. Immerse the S-A device in isopropyl alcohol in accordance with TT-I-735, Grade B, and observe for any indication of leakage. After removing assembly from the alcohol bath and drying, evacuate the assembly to 100 microns minimum. Maintain this vacuum for 5 minutes minimum, then pressurize the assembly to 25 psia using the fill gas specified above. Maintain this pressure for 5 minutes minimum. Crimp and cut off evacuation tube as shown in Drawing 917AS4000. Seal immediately by soldering to retain pressure. Repeat immersion test on evacuation tube seal within 3 minutes of crimping and sealing the evacuation tube.

(b) Within 4 hours after pressurizing and sealing the S-A device in accordance with 4.2.1.5(a), measure the leakage rate with a leak detector capable of measuring helium leakage at a rate of 5×10^{-6} standard cm^3/s or less.

4.2.2 End item quality conformance inspections. Unless otherwise indicated, the inspections specified herein shall be performed on the assembled, loaded, sealed, and pressurized S-A device to determine conformance with 3.2.1.2. For the purposes of this specification, the longitudinal axis and forward direction of the S-A device are defined on Drawing 917AS4000.

4.2.2.1 Drawing inspections. To determine conformance with 3.2.1.2.1, the following drawing inspections shall be performed on the S-A device. The item numbers assigned to the following inspections are a continuation of the item numbers used in 4.2.1.1.

CRITICAL - 100 percent inspection.

	REQUIREMENT	DRAWING
38.	Verify presence of safety inspection tag and lead seal (indicates safety determination has been made).	917AS4000

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	REQUIREMENT	DRAWING
MAJOR - Sample in accordance with 4.1.2.2.2:		
134.	Verify 0.218/0.211 inch diameter of through hole (five places).	917AS4023
135.	Verify true position of five 0.218/0.211 inch diameter holes.	917AS4023
136.	Verify 1.4375-18 UNEF-2B thread on booster end of housing.	917AS4023
137.	Verify that completely loaded units are identified by a band of yellow paint around exterior of housing.	917AS4000
138.	Verify presence, attachment, and content of identification plate (Drawing 917AS4028).	917AS4000
139.	Verify that protective cap NAS820-14A is installed.	917AS4000
140.	Verify presence of serial number on flange at location shown.	917AS4000
MINOR - None.		

4.2.2.2 Laboratory arming. The following conditions apply to performance of the laboratory arming tests to determine conformance with 3.2.1.2.2. Temperature conditioning and test temperatures shall be as specified in Figures 2 and 5 (see 4.2.2.2.3 for monitoring of switch functions during test).

(a) The arming characteristics of all S-A devices shall be monitored by the acoustic and TLM methods (4.2.2.2.1 and 4.2.2.2.2) simultaneously. Any S-A device which fails to meet the requirement of either method shall be considered defective.

(b) Step (e) of 4.2.2.2.1 and 4.2.2.2.2 shall not be performed in tests below the dashed line of Figure 2 nor in any of the tests of Figure 5.

(c) Only the results of the first attempt to pass each arming test shall be recorded for acceptance purposes.

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(d) For any arming test, the S-A device shall be mounted in a test fixture. The loaded fixture shall then be mounted on the centrifuge arm so that the flange end of the S-A device is toward the center of rotation and so that the specified acceleration will be applied at a point midway between the rotors. (The midpoint between rotors is located 5.591 inches from the aft face of the S-A device mounting flange.)

(e) During each arming test (except steps (a) of 4.2.2.2.1 and 4.2.2.2.2), after the specified acceleration level has been attained, a +28 ±5 Vdc pulse having a duration of 5 to 10 seconds shall be applied across pins D and H of the S-A device connector whenever necessary to actuate the launch latch solenoid.

(f) Except for those S-A devices which must be left in the armed condition for subsequent tests, all S-A devices shall be recycled to the safe condition in accordance with 4.2.2.2.4.1 after any test where arming takes place. Although failure to recycle will not cause a lot to be rejected, any individual S-A device which cannot be recycled will not be accepted by the procuring activity.

(g) At completion of arming tests, the condition of the S-A device (safe or armed) shall be determined visually through the viewing windows after removing solenoid voltage, but before removing the S-A device from the centrifuge.

4.2.2.2.1 Acoustic method. The acoustic method of measuring arming time can be accomplished by use of an acoustic pickup in conjunction with the preamplifier of Drawing 2238260 (or one having an equivalent output). NOLC Report 556 contains additional information on acoustic monitoring. The acoustic method of determining conformance with 3.2.1.2.2(a) through (e) consists of measuring the duration of rotation acoustically from the instant of applying solenoid activation voltage to the cessation of rotor movement. The S-A device shall be mounted on the centrifuge as specified in 4.2.2.2 and subjected to the following tests:

(a) With the launch latch solenoid actuated, subject the S-A device to a 3.00 ±0.03 gravity unit (g) acceleration in the forward direction along its longitudinal axis for 10 ±3 seconds. After acceleration has been terminated, examine the S-A device to determine whether arming has taken place, to determine conformance with 3.2.1.2.2(a).

NOTE. Care must be taken not to exceed the maximum acceleration level of this test at any time.

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(b) With the S-A device subjected to a 25.00 ±0.25 g acceleration in the forward direction along its longitudinal axis, energize the launch latch solenoid for at least 5 seconds, but not more than 10 seconds, and measure the time from application of solenoid voltage to cessation of rotor motion to determine conformance with 3.2.1.2.2(b).

(c) With the S-A device subjected to a 10.00 ±0.10 g acceleration in the forward direction along its longitudinal axis, together with a 20.00 ±0.20 g acceleration along an axis normal to its longitudinal axis, energize the launch latch solenoid for at least 5 seconds, but not more than 10 seconds, and measure the time from application of solenoid voltage to cessation of rotor motion to determine conformance with 3.2.1.2.2(c).

(d) With the S-A device subjected to a 6.00 ±0.06 g acceleration in the forward direction along its longitudinal axis, energize the launch latch solenoid for at least 5 seconds, but not more than 10 seconds, and measure the time from application of solenoid voltage to cessation of rotor motion to determine conformance with 3.2.1.2.2(d).

(e) With the S-A device subjected to a 5.00 ±0.05 g acceleration in the forward direction along its longitudinal axis, energize the launch latch solenoid for 10 seconds maximum. After acceleration has been terminated, examine the S-A device to determine whether arming has taken place (see 3.2.1.2.2(e)).

4.2.2.2.2 TLM method. The TLM method of determining conformance with the requirements of 3.2.1.2.2 consists of monitoring switches S2d and S2e. With the TLM points monitored, and with the S-A device mounted on the centrifuge as specified in 4.2.2.2, the following tests shall be performed:

(a) Perform the test of 4.2.2.2.1(a).

(b) Perform the test of 4.2.2.2.1(b), except that the time shall be measured from application of solenoid voltage to the closure of switch S2e.

(c) Perform the test of 4.2.2.2.1(c), except that the time shall be measured from application of the solenoid voltage to closure of switch S2e.

(d) Perform the test of 4.2.2.2.1(d), except that the time shall be measured from application of solenoid voltage to closure of switch S2e.

(e) Perform the test of 4.2.2.2.1(e).

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4.2.2.2.3 Switch functioning. During arming tests in accordance with 4.2.2.2, functioning of switches S2d and S2e shall be monitored at connector pins F and D with the control console monitor of LD 521549 (or equivalent), and with a recorder capable of indicating the opening and closing of each switch to determine conformance with 3.2.1.2.2.1.

4.2.2.2.4 Recycling and handling of armed S-A devices. After completion of the laboratory arming test of 4.2.2.2.1 or 4.2.2.2.2, the following procedures shall be used to recycle the S-A device to the safe position. After recycling, the unit shall meet the requirements of 3.2.1.2.6(b) (see Figures 2 and 5).

4.2.2.2.4.1 Recycling procedure. To recycle an armed S-A device to the safe condition, proceed as follows:

(a) Mount the armed S-A device in the test fixture of 4.2.2.2(d) with the forward (flange) end of the unit toward the center of rotation of the centrifuge.

(b) Apply an acceleration of at least 10 g and momentarily energize the launch latch.

(c) Increase acceleration to 60 ±5 g for at least 5 seconds. This action should lock the setback weight in overtravel position.

(d) Reduce acceleration to 10 to 15 g and index S-A device to a position at least 120 degrees from original position. Hold this acceleration for 8 ±2 seconds. This action should rotate the rotors back to safe position.

(e) Stop centrifuge.

(f) Energize launch latch solenoid, allowing the setback weight to return to the forward position. Deenergize the solenoid. A visual inspection through the S-A device viewing windows must show the following:

(1) Safe-arm indicator window. Letter "S" must be visible.

(2) Launch latch window. Latch must be engaged in the slot in the setback weight and there shall be no evidence of the red indicator as viewed through the window.

(g) If S-A device has not cycled to the safe condition, repeat steps (a) through (f).

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4.2.2.2.4.2 Handling of S-A devices that fail to recycle. Handle an armed S-A device that repeatedly fails to recycle as follows:

(a) Turn off all electrical power to centrifuge and S-A device.

(b) Wait for 15 minutes minimum before proceeding.

(c) Attach a connector to the S-A device which shorts pins C to D.

(d) Remove unit from fixture. Fill out and attach unsafe condition tag (Figure 7). This operation shall be performed by one person only. The S-A device shall be handled so that the output lead is facing away from personnel at all times.

(e) Place the S-A device in padded steel container and store in explosive area.

(f) Attach unsafe condition tag (Figure 7) to container.

(g) Disposition of failed S-A devices shall be at the discretion of the cognizant demolition officer (see 6.2.1).

4.2.2.3 Electrical resistance. To determine conformance with 3.2.1.2.3 perform the following inspections.

4.2.2.3.1 Resistance (safe condition). With the S-A device in the safe condition, measure the resistance across the pins listed in Table III (see 3.2.1.2.3.1) to determine conformance with 3.2.1.2.3.1.

WARNING

At no time shall the test current exceed 5 milliamperes (mA).

4.2.2.3.2 Resistance (armed condition). With the S-A device in the armed condition, measure the resistance across the pins listed in Table IV (see 3.2.1.2.3.2) to determine conformance with 3.2.1.2.3.2.

WARNING

To avoid danger of detonation, only test equipment which limits the test current to 5 mA shall be used in this test. Test points A to D and C to D are across the detonator circuit.

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CAUTION

To preclude the possibility of internal damage to the S-A device, do not apply negative voltage to pins A or B.

4.2.2.3.3 Detonator circuit continuity determination. Index the S-A device to a position at least 120 degrees from that of 4.2.2.2.4.1(a). Subject the S-A device to a 25.00 ±0.25 g longitudinal acceleration for a minimum of 15 seconds while monitoring test points C (-) to D (+) to determine conformance with 3.2.1.2.3.3.

WARNING

To avoid danger of detonation, only equipment which limits the test current to 5 mA maximum shall be used in this test. Test points C to D are across the detonator circuit.

4.2.2.4 Seal.

4.2.2.4.1 At place of manufacture. The sample units to be inspected at the contractor's facility for lot acceptance (Figure 4) shall be tested as specified in 4.2.1.5(b), except that the 4-hour time limit shall not apply.

4.2.2.4.2 After shipment. The sample units to be inspected after shipment from the contractor's facility (Figures 2 and 5) shall be subjected to both of the following tests:

(a) Immerse the S-A device in isopropyl alcohol in accordance with TT-I-735, Grade B, in a pressure vessel and reduce the pressure in the vessel to 10 inches of mercury absolute. Maintain the pressure for at least 5 minutes while observing for any indication of leakage.

(b) Test the sample as specified in 4.2.1.5(b), except that the 4-hour time limit shall not apply.

The S-A device shall meet the requirements of 3.2.1.2.4.2.

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4.2.2.5 Safety determination. After completion of arming-time tests and recycling, the contractor shall verify conformance with 3.2.1.2.5 as follows:

4.2.2.5.1 Visual verification.

(a) Observe position of arming rotor as viewed through the aft window to determine conformance with 3.2.1.2.5(a).

(b) Observe position of setback weight and launch latch, as viewed through the launch latch window, to determine conformance with 3.2.1.2.5(b).

4.2.2.5.2 Safety test. Following satisfactory verification of 4.2.2.5.1, the following test sequence shall be performed utilizing a test set that has been approved by the procuring activity. Observe that no explosive component of the S-A device initiates to determine conformance with 3.2.1.2.5(c).

CAUTION

For units in any condition other than safe, the firing train will initiate upon testing as specified below.

(a) Place the S-A device in an approved safety chamber, make electrical connections as necessary, and close the chamber cover.

(b) After placing the unit inside the safety chamber, apply +30 ±2 Vdc across pins H (+) and D (-) for 3 to 10 seconds.

(c) Apply +180 Vdc minimum across pins C (+) and D (-). This voltage must either be applied directly from a low impedance power supply, battery, or from a charged capacitor (minimum 0.33 microfarad (μF)). There shall be no resistors in the circuit limiting the current to the S-A device.

(d) A lead seal shall be crimped automatically after sensing the application of the +180 Vdc described above. This lead seal, embossed with an identifiable symbol of the testing agency, shall fasten the safety-inspection tag (Figure 1) to the S-A device.

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(e) Immediately following this test, the S-A device shall be packaged for shipment.

Prior to any subsequent testing, the safety-inspection tag shall be removed from the unit.

4.2.2.6 Visual inspection. To determine conformance with 3.2.1.2.6, prior to packaging, the S-A device shall be inspected for external damage, verification that the letter "S" is visible in the aft window and no red is visible in the forward window, verification that a safety-inspection tag is attached to housing, lead seal is crimped onto safety-inspection tag wire, safety-inspection tag is filled in and signed, a 3/4-inch-wide yellow band has been painted around the housing, and that workmanship meets the requirements of 3.2.1.2.6(h) and 3.3.3.

4.2.2.7 Environmental tests. The preproduction sample, or a randomly selected sample of 15 units from an inspection lot, shall be subjected to the tests of this paragraph in accordance with Figures 2 or 5, as applicable. Any S-A device subjected to these tests shall be considered destroyed and shall not be returned to the lot nor forwarded for service use, but shall be labeled appropriately and held for disposal instructions from the procuring activity (see 6.2.1). Initiation of any explosive component, except for those which are detonated intentionally as part of a test or by test error, shall cause rejection of the lot from which the unit was drawn. For vibration testing, the test fixture shall be secured to the vibration table with a torque sufficient to prevent loosening during vibration. The S-A device shall be attached to the required test fixture, using four MS16997-33 capscrews (8-32 x 5/8 inch) and four MS35333-38 washers; the screws shall be tightened to a torque of 25 ±2 inch-pounds. Throughout the environmental tests, the S-A device shall be in the safe condition unless otherwise indicated. (See 4.1.2.2.3 for inspection lot requirements.)

WARNING

The explosive components of this S-A device are highly sensitive and hazardous; therefore, the S-A device must be handled with extreme care. An inert Booster, Fuze, MARK 38 MOD 1 (LD 521540) shall be used with the S-A device during the tests of 4.2.2.7.2, 4.2.2.7.5, and 4.2.2.7.6. The booster pellet (Drawing 1778086) shall be of an inert material of equivalent density to that of the original.

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4.2.2.7.1 Jolt. To determine conformance with 3.2.1.2.7.1, the S-A device shall be subjected to MIL-STD-331, Test 101, Procedure II. Each unit shall be mounted for testing in the test fixture of Drawing SA 2602010. Following the jolt test, the S-A device shall be tested in accordance with 4.2.2.2.1(b) and 4.2.2.2.2(b) to determine conformance with 3.2.1.2.2(b). Disassembly and inspection after the test shall be to the extent required to determine conformance with the requirements of MIL-STD-331.

4.2.2.7.2 Forty-foot drop. To determine conformance with 3.2.1.2.7.2, the S-A device shall be subjected to MIL-STD-331, Test 103, Procedure 1, except that the S-A device shall be dropped in the warhead fixture of Drawing SA 2618609. The S-A devices to be expended in this test shall be dropped in the positions described in Figures 2 and 5. Disassembly and inspection after the test shall be to the extent required to determine conformance with 3.2.1.2.7.2.

4.2.2.7.3 Transportation vibration. To determine conformance with 3.2.1.2.7.3, the S-A device shall be subjected to MIL-STD-331, Test 401, Procedure II. Temperature conditioning and test temperatures shall be as specified in Figure 2 or 5. The S-A device shall be mounted for testing on the fixture of Drawing 2408846, then the fixture shall be mounted on a vibration table as specified in 4.2.2.7.

4.2.2.7.4 Humidity. To determine conformance with 3.2.1.2.7.4, the S-A device shall be subjected to the test of MIL-STD-810, Method 507, except the test shall be conducted as follows:

> The S-A device shall be placed in a temperature and humidity chamber at 16 ±6 degrees Celsius (°C) and 95 ±5 percent relative humidity. The temperature and humidity shall be changed over a 1-hour period to 60 +0°C, -6°C and 70 ±20 percent relative humidity and maintained for 3 hours. Over a 1-hour period, the temperature and humidity shall be changed to 16 ±6°C and 95 ±5 percent relative humidity and maintained for 3 hours. This procedure shall constitute one cycle and shall be repeated for a total of 26 cycles. The S-A device may be stored over weekend periods at 23 ±10°C and 50 ±30 percent relative humidity.

The electrical connector of the S-A device shall be thoroughly cleaned and dried prior to performing subsequent electrical tests.

4.2.2.7.5 Flight vibration before arming. To determine conformance with 3.2.1.2.7.5, the S-A device shall be subjected to 2 hours of random vibration having a spectral density of 0.04 gravity units squared per hertz (g^2/Hz) ±3.0 decibels (dB) over a bandwidth of 20 to 2,000 hertz (Hz) with a minimum rolloff of 6 dB per octave applied along each of three mutually perpendicular axes, one

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of which shall be the longitudinal axis, for a total time of 6 hours. Vibration amplitude peaks may be limited to three times the average amplitude, if required, to protect the test equipment. For this test, the test fixture of Drawing 2408846 shall be used. The S-A device and test fixture shall be mounted on a vibration table as specified in 4.2.2.7. Temperature conditioning and test temperatures shall be as specified in Figure 2 or 5.

4.2.2.7.6 Flight vibration after arming. To determine conformance with 3.2.1.2.7.6, the S-A device, in the armed condition with the launch latch solenoid energized, shall be subjected to 5 minutes of random vibration along each of the three major axes for a total time of 15 minutes. The spectral density shall be 0.04 g^2/Hz ±3.0 dB over a bandwidth of 50 to 2,000 Hz with a minimum rolloff of 6 dB per octave. Vibration amplitude peaks may be limited to three times the average amplitude, if required, to protect the test equipment. For this test, the test fixture of Drawing 2408846 shall be used. The S-A device and test fixture shall be mounted on a vibration table as specified in 4.2.2.7.

4.2.2.8 Explosive train tests. The power source used to fire the detonators during the static detonator safety test (4.2.2.8.1) and the explosive train output test (4.2.2.8.2) shall consist of the following components:

(a) A power source with an output of +180 ±18 Vdc.

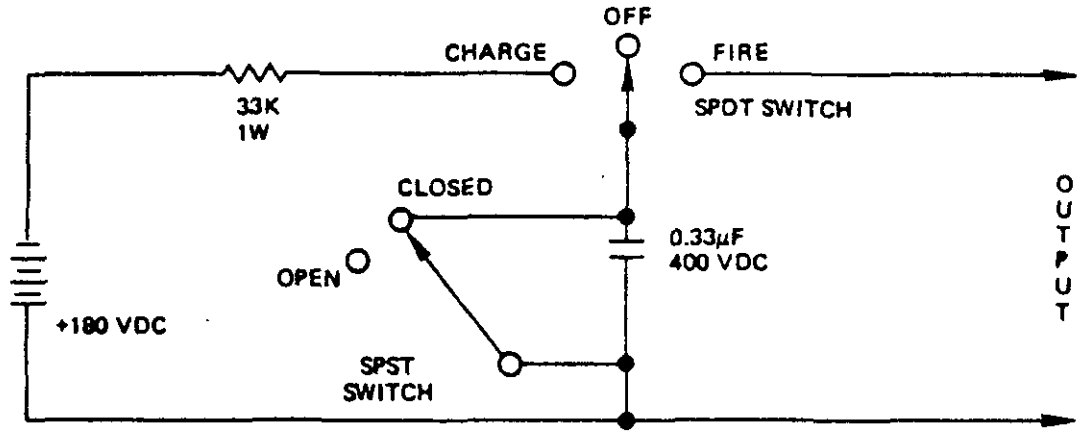
(b) A 33.0 ±3.3 kohm, 1 watt resistor.

(c) A 0.33 ±0.03 µF, 400 Vdc capacitor.

(d) A single-pole double-throw (SPDT) switch, spring-loaded to the OFF position.

(e) A single-pole single-throw (SPST) switch, spring-loaded to the CLOSED position.

The components shall be arranged as follows:



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4.2.2.8.1 Static detonator safety. The following internal circuit modifications must be made to accomplish detonator firing with the S-A device in the unarmed condition. To determine conformance with 3.2.1.2.8.1, proceed as follows:

WARNING

Proper safety precautions must be observed when making the following modifications, and care must be taken to prevent accidental initiation of the detonators by static potentials.

(a) Establish a shorting circuit from pins C to D and from A to D before removing mechanism assembly from housing.

(b) Remove mechanism from housing.

(c) Solder an insulated jumper from terminals 12 to 7 and from 1 to 11 on the mechanism circuit board to close the firing circuit.

(d) Wrap S-A device housing with tape. (See note at end of paragraph.)

(e) Remove rotor switch.

(f) Replace the mechanism in its housing. The internal circuit is now electrically armed and the detonators are ready to fire when the shorting circuit is removed.

(g) After proper precautions have been taken to ensure the safety of personnel, remove shorting circuit from pins C to D and A to D. Connect power source specified in 4.2.2.8 to pins C and D.

WARNING

The S-A device must be installed in approved safety chamber prior to performance of step (h).

(h) Fire S-A device as follows:

(1) Place and hold SPST switch in OPEN position.

(2) Place SPDT switch in CHARGE position for 10 seconds minimum, then place in FIRE position.

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(3) Return SPDT switch to OFF position.

(4) Release SPST switch to CLOSED position.

NOTE. To aid in determining whether the velocity of ejected parts is sufficient to cause a damaging effect when the S-A device is fired, the S-A device housing shall be wrapped with two layers of polyester pressure-sensitive tape, Drawing 917AS4087, centered over each window. Ends of the tape may overlap not more than 3/4 inch, but the overlap must not be over either window. After detonator initiation, the tape shall be examined for rupture.

4.2.2.8.2 Explosive train output. To determine conformance with 3.2.1.2.8.2, the detonators of the S-A device shall be fired with the S-A device in the armed condition. Proceed as follows:

(a) Install a dent block (Drawing SA 2602266) on the threaded end of the S-A device.

WARNING

The S-A device must be installed in an approved safety chamber prior to performance of step (b).

(b) After clearing the area, initiate the explosive train of the S-A device, using the firing circuit of 4.2.2.8.

(c) Remove any foreign material from the dent and measure the depth of dent. The depth of dent shall be determined by taking the average of two readings, one each at two points 1/8 inch in from opposite edges of the dent block as compared to the deepest part of the dent.

4.3 Packaging inspection. Each package and shipping crate shall be inspected to determine conformance with Section 5.

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5. PACKAGING (see 6.2.1).

5.1 Preservation-packaging.

5.1.1 Level A. Seven S-A devices per container shall be packaged in accordance with DL 917AS3109. Each container shall have the inspection record of Figure 6, one for each S-A device, placed on top of contents before sealing.

5.1.2 Level C. Not applicable, unless specified in the contract or purchase order.

5.2 Packing.

5.2.1 Level C. The containers of 5.1.1 shall be packaged in accordance with normal commercial practice, and shall be protected against damage during shipping, storage, and handling. Packing shall conform to the Code of Federal Regulations, 49 CFR 171-179.

5.3 Marking.

5.3.1 Packages. Unless otherwise specified in the contract or purchase order, each package shall be marked in accordance with DL 917AS3109.

5.3.2 Shipping containers. Unless otherwise specified in the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 and the Code of Federal Regulations, 49 CFR 171-179.

6. NOTES.

6.1 Intended use. The MARK 33 MOD 1 safety-arming device is a component of the Sparrow AIM-7F guided missile fuzing system.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Number, title, and date of this specification.

(b) Government-furnished property list, if required (see 3.1.1).

(c) When a preproduction sample is required, testing and approval will be under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulations. The preproduction sample should consist of 45 S-A devices. The contracting officer should include specific

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instructions in all procurement instruments regarding arrangements for inspection, test, and approval of the preproduction sample. The necessity for a preproduction sample shall be specified when production under a new contract, by the same contractor at the same location, follows the manufacture of satisfactory S-A devices as covered by this specification (see 3.4.1 and 4.1.2.1).

(d) Responsibility for inspection (see 4.1.1).

(e) A facility to perform the inspections and tests of Figure 5 (see 4.1.2.2(b)).

(f) Inspection lot, if different from 4.1.2.2.3.

(g) Serial numbers (see 4.1.2.2.3).

(h) A provision that, following rejection of three successive inspection lots, the procuring activity will not accept samples from additional lots for testing until the cause or causes for failure have been resolved and remedied to the satisfaction of the procuring activity.

(i) Inspection and acceptance records (see Figure 6) are to include the results of all tests and inspections specified herein and any additional tests, inspections, repairs, rework, replacements, or retests of any part, subassembly, assembly, or end item deemed necessary by the contractor to achieve selection or acceptance. These records are to be maintained by the contractor and must be available to the procuring activity for review (see 4.1.2.3).

(j) Instructions for the disposition of units or lots that are destroyed or rejected (see 4.1.2.2.1, 4.2.2.2.4.2, and 4.2.2.7). The following instructions are recommended (see 6.3 and 6.4 for reworked components): "Rejected units or lots may be resubmitted at the discretion of the procuring activity, provided the rework plan or waiver request is approved by the Government. The procuring activity will not accept a unit or a lot for testing following the third rejection of the unit or lot. Records of any units which have been reworked shall be maintained by the contractor for a period of 5 years after contract completion. Records of units which have been destroyed by testing, or scrapped, shall be maintained for a period of 1 year."

(k) Level of packaging, packing, and marking (see section 5).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions

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of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.1.2.3	Inspection record	DI-T-2072
4.1.2.3	Lot inspection report	DI-T-2072
4.1.2.4	Production summary reports	DI-P-3475A

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Disposition of defective units within lots accepted for further testing. Defective units within lots accepted for further testing may be reworked and resubmitted to the test sequence for individual acceptance. However, those units which are classified as defective after the third submission will be scrapped. These units will be destroyed in their entirety, and no parts therefrom will be used for further production.

6.4 Disposition of preproduction or lot samples. At the discretion of the procuring activity, if the preproduction sample or lot sample fail to meet the acceptance criteria of 4.1.2.1.1 or 4.1.2.2.1, as applicable, components thereof may be reworked, resampled, and resubmitted to the test sequence for preproduction sample or lot acceptance. However, those lots which are rejected after the third submission will be scrapped. All units within those lots will be destroyed in their entirety, and no parts therefrom will be used for further production.

6.5 Safety precautions. The manufacture of this item in accordance with this specification requires loading and handling of explosive material during assembly. These operations are hazardous to personnel and property and, therefore, require suitable explosive safety precautions. Safe practices and standards for all work performed in connection with Department of Defense (DoD) contracts involving ammunition, explosives, and related dangerous material are set forth in DoD 4145.26M.

6.6 Definitions.

6.6.1 Certification. Actual test and inspection results or other reliable and verifiable quality data.

Preparing activity :
Navy (AS)
(Project No. 1336-N327

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(FRONT)

SAFETY-INSPECTION TAG

S-A DEVICE MARK 33 MOD 1 S/N _____ Lot No. ____

This unit was inspected and found to be in the SAFE condition on the date indicated (silver letter "S" on green background visible in aft window, launch latch fully engaged (forward window), and no evidence of red indicator as viewed through either window).

Signature	Date
_____ Manufacturing Co. Rep[a]	_____
_____ Government Inspector	_____
_____ Dept Recv Inspector	_____
_____ Installing Personnel	_____

[a]Contractor's name and signature

(BACK)

This tag shall be placed in the missile folder when unit is installed in the missile.

CAUTION: If at a depot or any using facility, this S-A device is found to be in an armed condition, or if it cannot be determined to be in a SAFE condition, it shall be inspected and verified as to its condition-- either armed or indeterminate-- by an ordnance officer and disposed or in accordance with existing regulations for explosive material. A report of the incident, including serial number, Mark and Mod designation, date of inspection and any known history of environmental exposure shall be forwarded to the Naval Air Systems Command with this tag.

This tag, including material for attachment, shall meet the requirements of UU-T-81, Type B, Grade 15 CSU, Size 7 or 8, White. The information may be stamped or printed thereon.

FIGURE 1. Safety-inspection tag.

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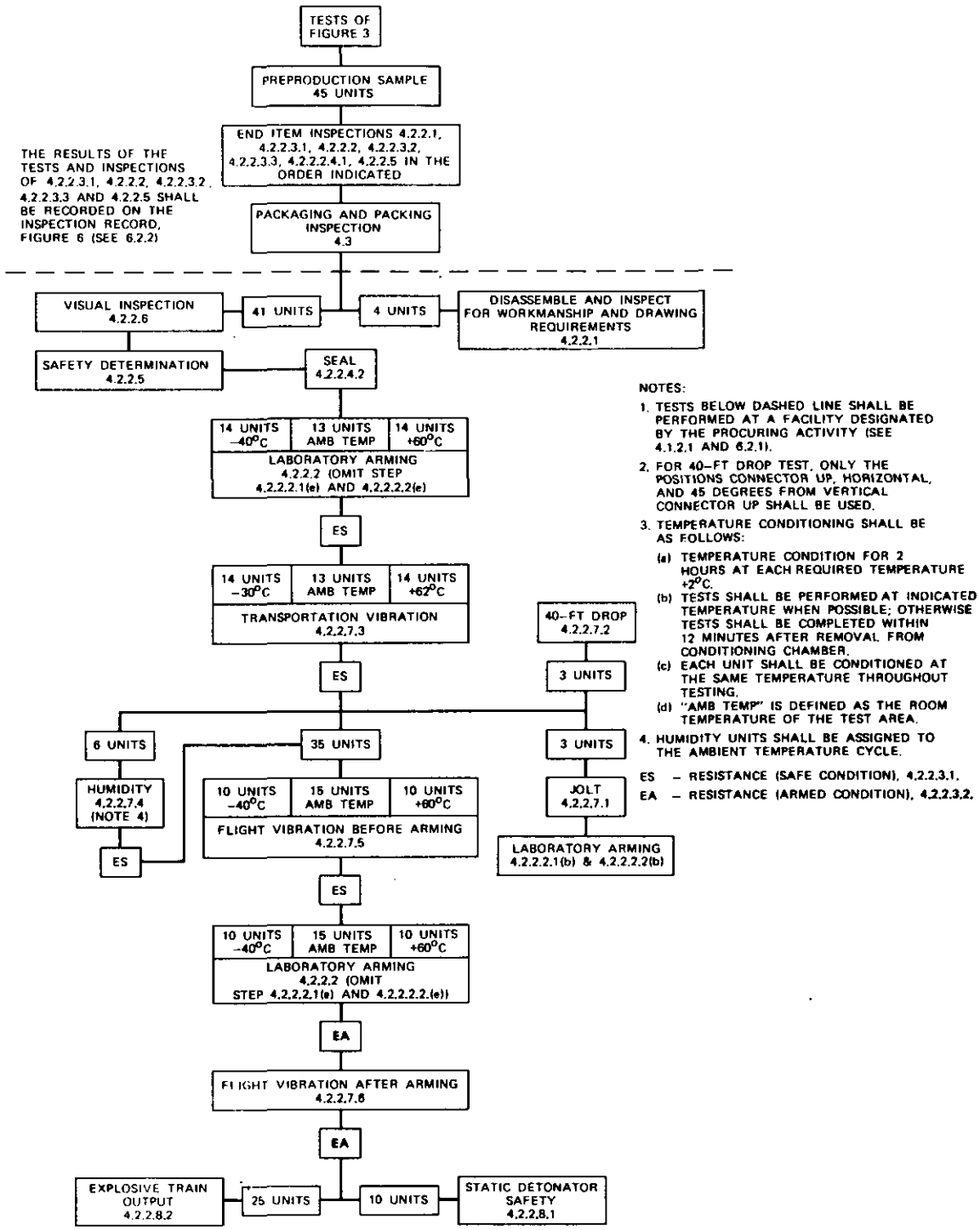
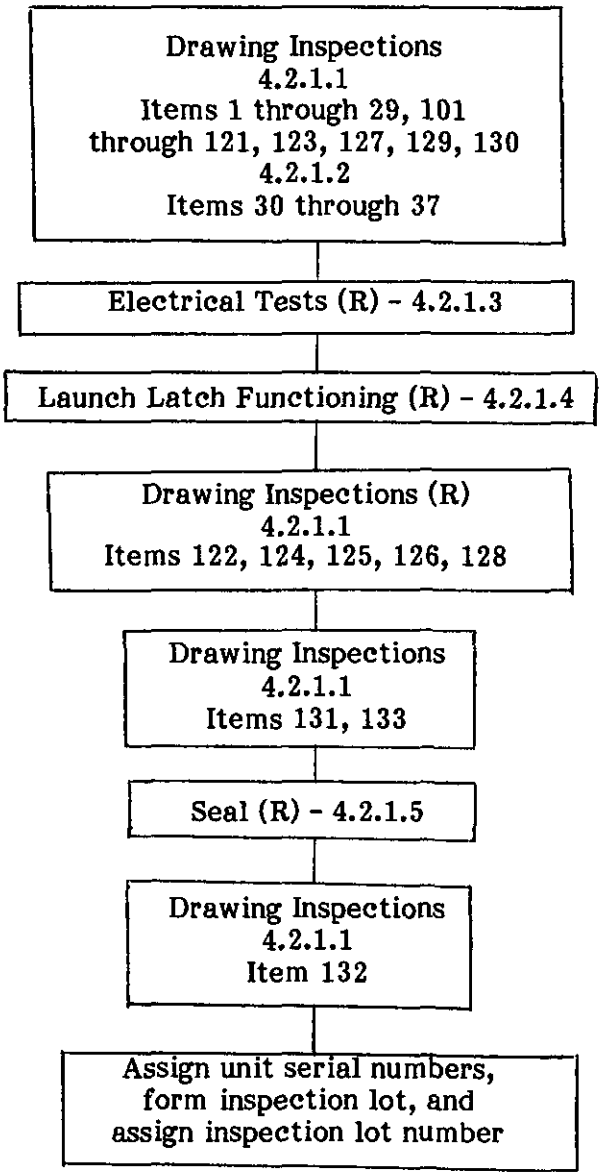


FIGURE 2. Preproduction sample test sequence.

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(R) = Record results in inspection record, Figure 6.

FIGURE 3. In-process inspection sequence.

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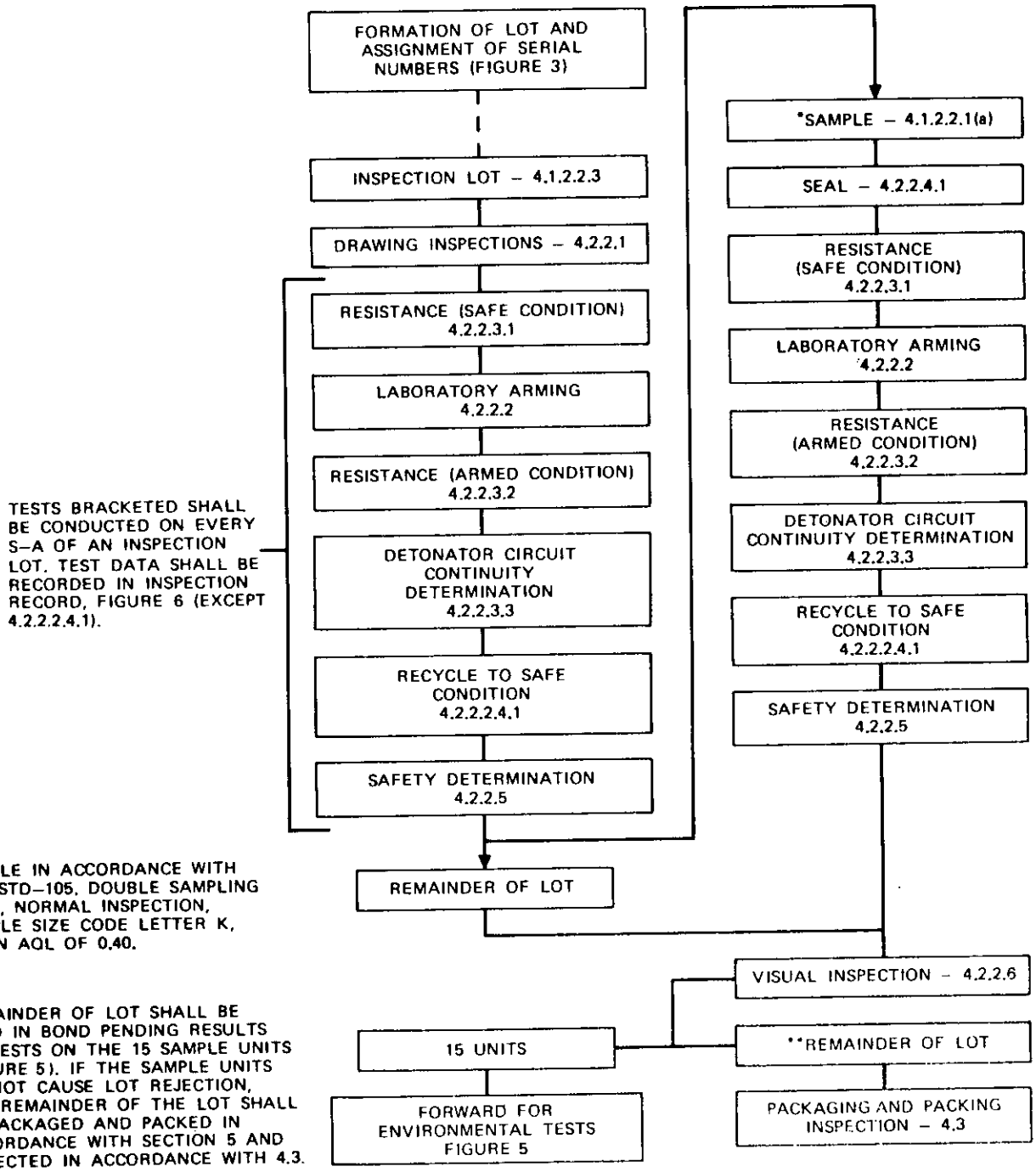


FIGURE 4. End-item inspection sequence (nondestructive tests).

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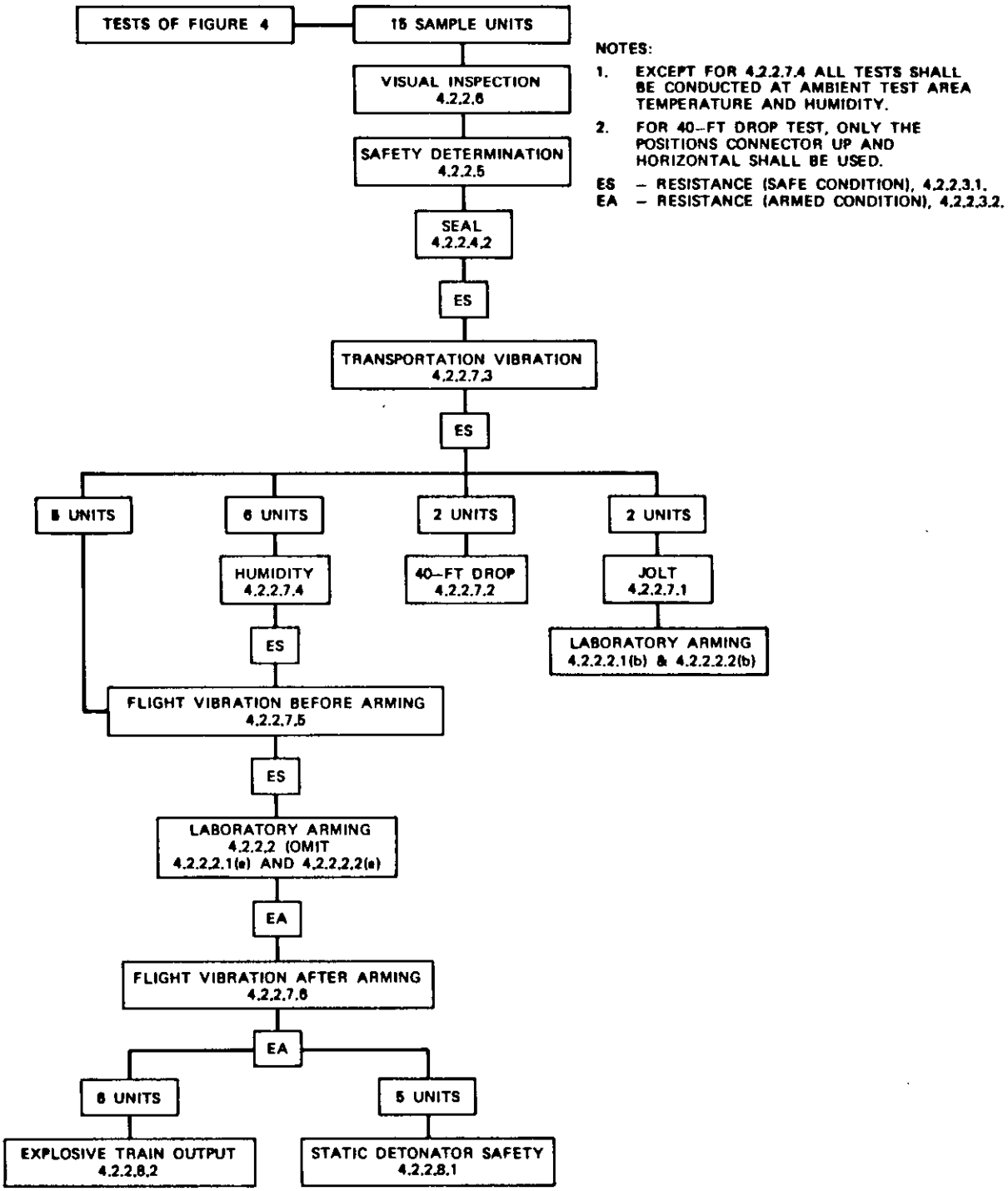


FIGURE 5. Environmental (destructive) test sequence (including pre-environmental and post-environmental tests).

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INSPECTION RECORD

S-A DEVICE MARK 33 MOD 1

CONTRACT NO. _____

SERIAL NO. _____

LOT NO. _____

I. ELECTRICAL TESTS (4.2.1.3)

Test No.	Limits		Reading	Unit
	Minimum	Maximum		
1	100	. . .	_____	Mohm
2	100	. . .	_____	Mohm
3	100	. . .	_____	Mohm
4	100	. . .	_____	Mohm
5	100	. . .	_____	Mohm
6	100	. . .	_____	Mohm
7	10.0	12.5	_____	kohm
8	27.0	32.0	_____	ohm
9	1.8	2.2	_____	kohm
10	13.5	16.5	_____	kohm
11	4.0	11.0	_____	Mohm
12	16.2	. . .	_____	µF
13	. . .	0.15	_____	ohm
14	. . .	0.15	_____	ohm
15	. . .	0.30	_____	ohm
16	19.30	24.40	_____	kohm
17	3.75	4.59	_____	kohm

Contractor's Inspector _____

FIGURE 6. Inspection record.

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II. LAUNCH LATCH FUNCTIONING (4.2.1.4)

A. Yes _____ No _____
B. Yes _____ No _____

Contractor's Inspector _____

III. LOADING (4.2.1.1, items 122, 124, 125, 126, 128)

A. 2 detonators, electric, MARK 71 MOD 0 Yes ___ No ___
B. 1 lead, explosive, MARK 8 MOD 0 Yes ___ No ___
C. 1 lead, explosive, MARK 16 MOD 0 Yes ___ No ___

Contractor's Inspector _____ Government Inspector _____

IV. SEAL (4.2.1.5) Passed _____ Failed _____

Contractor's Inspector _____ Government Inspector _____

V. LABORATORY ARMING (4.2.2.2)

A. Arming Time	4.2.2.2.1 value	4.2.2.2.2 value
(a) Passed___ Failed___	Not applicable	Not applicable
(b) Passed___ Failed___	_____sec.	_____sec.
(c) Passed___ Failed___	_____sec.	_____sec.
(d) Passed___ Failed___	_____sec.	_____sec.
(e) Passed___ Failed___	Not applicable	Not applicable

B. Switch Functioning (4.2.2.2.3)

S2d conforms to 3.2.1.2.2.1 requirements Passed___ Failed___
S2e conforms to 3.2.1.2.2.1 requirements Passed___ Failed___

Contractor's Inspector _____ Government Inspector _____

FIGURE 6. Inspection record (contd).

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VI. ELECTRICAL RESISTANCE (4.2.2.3)

RESISTANCE (SAFE CONDITION) (4.2.2.3.1)

Test No.	Resistance limits		Reading	Unit
	Minimum	Maximum		
* 1	100	. . .	_____	Mohm
* 2	100	. . .	_____	Mohm
* 3	100	. . .	_____	Mohm
4	10.0	12.5	_____	kohm
5	27.0	32.0	_____	ohm
6	13.5	16.5	_____	kohm
7	4.0	11.0	_____	kohm

[a]If test results are below minimum specified with D connected to frame on negative side, then check frame to C and D to C individually to verify if resistance is within limits specified.

* = Resistance shall be met with 500 ±50 Vdc applied

Contractor's Inspector _____ Government Inspector _____

RESISTANCE (ARMED CONDITION) (4.2.2.3.2)

Test No.	Resistance limits		Reading	Unit
	Minimum	Maximum		
1	1.0	4.0	_____	ohm
2	27.0	32.0	_____	ohm
3	3.8	4.6	_____	kohm

Contractor's Inspector _____ Government Inspector _____

FIGURE 6. Inspection record (contd).

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VII. DETONATOR CIRCUIT CONTINUITY DETERMINATION (4.2.2.3.3)

Passed _____ Failed _____

Contractor's Inspection _____ Government Inspector _____

VIII. SAFETY DETERMINATION (4.2.2.5)

Visual Verification

4.2.2.5.1(a) Passed _____ Failed _____
4.2.2.5.1(b) Passed _____ Failed _____

Safety Test

4.2.2.5.2 Passed _____ Failed _____

Contractor's Inspection _____ Government Inspector _____

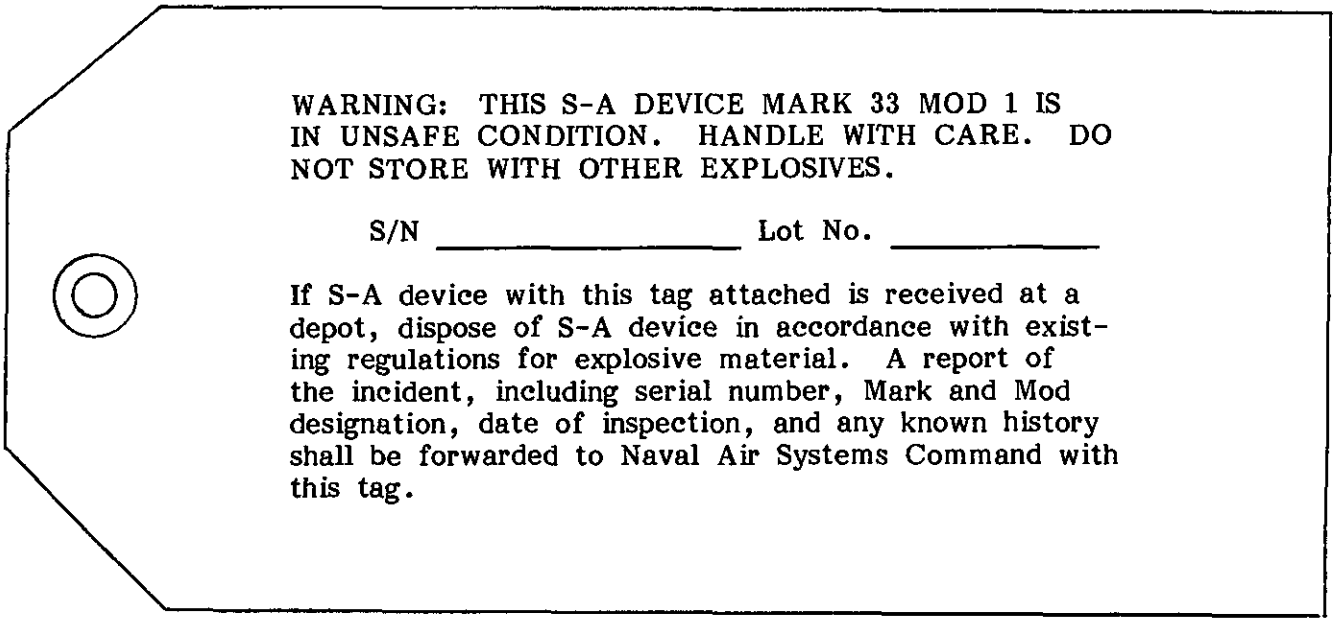
IX. VISUAL INSPECTION (4.2.2.6)

(1)	External Damage	Yes ____	No ____
(2)	Launch latch	Yes ____	No ____
	(a) "S" visible aft window	Yes ____	No ____
	(b) No red forward window	Yes ____	No ____
(3)	Yellow band housing	Yes ____	No ____
(4)	Dust cap installed	Yes ____	No ____
(5)	Safety tag attached and properly filled out	Yes ____	No ____
(6)	Lead seal crimped onto safety inspection tag wires	Yes ____	No ____
(7)	All signatures on safety tag as required	Yes ____	No ____
(8)	Workmanship	Passed ____	Failed ____

Contractor's Inspector _____ Government Inspector _____

FIGURE 6. Inspection record (contd).

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This tag, including material for attachment, shall meet the requirements of UU-T-81, Type B, Grade 15 CSU, Size 7 or 8, Red. The information may be stamped or printed thereon.

FIGURE 7. *Unsafe condition tag*.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-85333(AS) - Safety-Arming Device MARK 33 MOD 1

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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A. GIVE PARAGRAPH NUMBER AND WORDING

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C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

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DATE

DD FORM 1426 1 OCT 76

Replaces edition of 1 Jan 72 which may be used.

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